

PhysRNet: A Physics-Aware Reasoning Network with Exact E(2)-Equivariance, Conservation Laws, and Interpretable Interaction Graphs

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Abstract—We present PhysRNet, a three-module architecture for learning physical force laws from trajectory data. The architecture combines (i) an E(2)-equivariant encoder with exact rotation-equivariance by construction (error $\leq 3 \times 10^{-12}$), (ii) physics-informed conservation losses (energy, momentum, angular momentum) that reduce long-horizon drift by 30%, and (iii) a graph-based reasoning module that predicts interaction strengths as learned edge weights. A per-node gate blends the two predictive pathways while preserving exact equivariance. PhysRNet learns three conservative force laws—Plummer gravity, Hooke springs, Lennard-Jones—from synthetic trajectories alone, generalises to unseen masses and configurations, and runs on CPU in under three minutes per system.

Index Terms—equivariant neural networks, physics-informed networks, graph neural networks, force-law learning, conservation laws

I. INTRODUCTION

Learning physical force laws from trajectory data requires correct symmetry structure and conservation constraints. We present **PhysRNet**, combining three proven ideas:

- 1) **Exact E(2)-equivariance.** Every scalar coefficient in the architecture is a function of rotation-invariant features only; every vector is a scalar $\times$ equivariant vector. The full model is exactly equivariant, verified to machine precision.
- 2) **Conservation constraints.** Auxiliary losses penalise violations of energy, linear, and angular momentum conservation.
- 3) **Interpretable interaction graph.** The reasoning module predicts signed edge weights w_{ij} such that $\mathbf{a}_i = \sum_j w_{ij} \hat{\mathbf{u}}_{ij}$.

A per-node learned gate $g_i \in [0, 1]$ blends the equivariant and reasoning pathways while preserving exact equivariance.

II. ARCHITECTURE

A. Node features

Particle states $(\mathbf{r}_i, \mathbf{v}_i, m_i)$ are encoded as:

- **Invariant scalars:** $\mathbf{s}_i = [m_i, |\mathbf{v}_i|, \frac{1}{2}m_i|\mathbf{v}_i|^2, |\mathbf{r}_i - \bar{\mathbf{r}}|] \in \mathbb{R}^4$.

- **Equivariant vectors:** centroid-centred position and velocity, padded to 3D, as 2 vector channels.

B. Equivariant encoder

$L = 2$ rounds of message passing over a fully-connected graph. Edge features are rotation-invariant ($\mathbf{s}_i, \mathbf{s}_j$, distance norms, dot products). Scalar messages are MLPs of invariant features. Vector messages are $\sigma(\text{MLP}) \times (\mathbf{v}_i - \mathbf{v}_j)$ —scalar gate $\times$ equivariant carrier. Node updates multiply aggregated vector messages by invariant scalar gates.

C. Equivariant acceleration head

From the encoder’s equivariant vector features, a linear head with invariant scalar coefficients produces per-node acceleration: $\mathbf{a}_i^{\text{equiv}} = \sum_c c_i \cdot \mathbf{v}_{i,c}^{\text{enc}}|_{2\text{D}}$.

D. Reasoning GNN

The reasoning module computes per-pair interaction weights: $w_{ij} = \text{MLP}([\mathbf{s}_i, \mathbf{s}_j, d_{ij}, |\Delta \mathbf{v}|])$, and predicts acceleration as $\mathbf{a}_i^{\text{reason}} = \sum_j w_{ij} \hat{\mathbf{u}}_{ij}$. The learned edge weights w_{ij} are the recovered interaction graph.

E. Physics residual

Conservation-law losses from semi-implicit Euler prediction:

$$\mathcal{L}_{\text{phys}} = |E_{\text{pred}} - E_{\text{true}}|^2 + 0.5|\mathbf{p}_{\text{pred}} - \mathbf{p}_{\text{true}}|^2 + 0.5|\mathbf{L}_{\text{pred}} - \mathbf{L}_{\text{true}}|^2 \quad (1)$$

F. Training

Loss: $\mathcal{L} = \|a_{\text{pred}} - a_{\text{true}}\|^2 + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}}$, $\lambda_{\text{phys}} = 0.5$. Adam (2×10^{-3}), cosine annealing, batch 512, 60 epochs.

III. EXPERIMENTS

All experiments on CPU, 300 training trajectories, 50 timesteps, hidden dim 64.

Force-law learning. PhysRNet learns all three force laws from trajectories alone (Table I). Gravity achieves the lowest

System	Test MSE	Gen MSE	Equiv. err	Drift
Gravity	4.1×10^{-4}	5.9×10^{-4}	6.4×10^{-14}	0.003
Spring	3.4×10^{-2}	1.1×10^{-2}	1.6×10^{-12}	0.015
LJ	4.6×10^{-2}	1.2×10^{-2}	2.6×10^{-13}	0.019

TABLE I

TEST AND GENERALISATION MSE, ROTATION EQUIVARIANCE ERROR, AND 50-STEP ENERGY DRIFT ON THREE CONSERVATIVE SYSTEMS.

MSE (4.1×10^{-4}); spring and LJ converge to $\sim 10^{-2}$ within 60 epochs.

Exact equivariance. The rotation error $\|R \cdot a(x) - a(Rx)\|^2$ is at machine precision ($\leq 3 \times 10^{-12}$) for all systems and all model variants. This is an architectural guarantee, not an empirical property.

PINN ablation. On gravity, removing the physics loss increases MSE by $\times 2.2$ and energy drift by $\times 1.4$, confirming that conservation constraints act as effective regularisers.

Reasoning-only. Freezing the gate ($g = 0$) shows the reasoning module alone achieves moderate MSE (spring: 0.19 versus 0.034 for the full model), proving the interpretable pathway learns the force law independently.

IV. CONCLUSION

PhysRNet combines exact symmetry, conservation, and interpretability into a single architecture for physical force-law learning. The exact equivariance guarantee is verified to machine precision, the conservation loss reduces drift by 30%, and the reasoning module recovers interaction strengths as explicit edge weights. All results are CPU-reproducible from committed code.

REFERENCES